

Effects of collision warning system under different warning timing on driving speed and distance

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Abstract—The study examined impacts of different releasing time of warning with an intersection collision warning system (ICWS) on driver behavior. Forty five participants drove a simulated vehicle through five driving scenes containing seven different unexpected collision events with different releasing time of warning and no warning (control) group was presented to counter balanced with the collision event types. Comparing with no warning group, speed trends had significant differences under different releasing time of warning conditions in this study. The results also indicated that releasing ICWS warnings could reduce driving speed to a hazard event so that drivers would adjust speed more appropriately to respond to collision situation. Considering the time of application of violent deceleration for different releasing time of warning, 3.0 s to 4.0 s before a collision occurred might be a proper time to provide warning. In terms of distance to conflict point with maximum deceleration, different releasing time of warning showed statistically different, which suggested that drivers would reserve more space to clear impending collision with earlier warning. Additionally, with the time of warning released earlier, the time of the maximum deceleration rate appeared earlier. Better designs of effective warning when driving at critical intersections could be derived from these results.

Keywords—Collision warning system; Auditory warning; Releasing time of warning; Driving speed;

I. INTRODUCTION

Safe driving at intersections was one of the most dynamic and complex tasks for drivers on road, and a high degree of visual and spatial processing was required [1-2] in this task. According to in-depth analyses of accidents, recognition error was one of the main contributing factors for accidents at intersections [3], which had found that lack of information

was the main reason leading to an accident in above 90% of intersection accidents. The similar result was showed in the research [4]. In most cases, the other traffic participants on the road were not noticed by the drivers and they even failed to yield to the other road users with priority.. Thus, in-vehicle warning information systems were installed for the purpose of enhancing drivers' safety and it could be also used to help drivers deal with the complex situation at intersections.

Due to advances in in-vehicle collision warning system technology, driver's safety benefits a lot from the warnings. Several simulator and on-road studies suggested that the warning could produce several potential benefits, such as reducing reaction time to hazardous situations [5-8], reducing collision involvements[9-11]. Additionally, it was well-known that the effectiveness of collision warning system warnings was affected by various parameters [12-16]. A key, of course, was that the released warnings information must improve collision warning system effectively if they are to be useful. A study [17] had indicated that the auditory warning information could assist drivers in detecting the RLR vehicles in a timely manner, and early warning provided drivers more adequate time and space to decelerate to avoid collisions with the conflicting vehicle.

The studies mentioned above indicated that the drivers could greatly benefit from the auditory warning on the direction of the risk in driving. To date, there had been few promising studies on analyzing the effect of different warning releasing time on drivers' behavior [18-19]. Therefore, more research were needed to further determine the proper releasing time. Experiments were conducted to evaluate the effect of different warning releasing time on driving performance including driving speed trend and distance to conflict point.

And the results obtained could contribute to a better design of warning content for in-vehicle warning systems, which would in turn enhance road safety and efficiency.

II. METHODS

A. Participants and apparatus

A total of 45 subjects (53% males) participated in the study. Participants ranged from 31 to 39 years of age (mean age 35 years). With regard to their driving experience, all participants had a driver's license for at least 3 years and drove more than twenty thousand kilometers per year. All participants had normal or corrected to normal vision and hearing.

In this study, BJTU Driving Simulator developed by Real Time Company was used to present the simulated driving scenarios and ICWS warnings. The simulator integrates a real vehicle (a Ford Focus), a virtual reality-based visual and audio system, one degree of freedom motion platform which can imitate the feeling of motion and a host computer system to simulate the virtual environment of an urban road. Inside the driving cabin, the simulator is equipped with full operational control including a steering wheel, brake and accelerator pedal as well as turn signal indicators, etc. Also, the driving simulator has a 300-degree front view of display system with simulated back mirrors. Side-view screens and a rear-view screen allow the driver to monitor traffic from all directions.

B. Driving scenario and Experiment design

In order to examine the effectiveness of the ICWS warnings with different releasing time in a critical intersection, a situation with a high collision probability was required. Studies had mentioned that red light running crashes were found to be more severe than other types of crashes, and other accidents studies had shown similar results [20-21]. Thus, a Red-light-running (RLR) collision scenario in this experiment was defined as conflict event at an experimental intersection. When drivers entered the experimental intersection, the traffic light for simulator was green, meanwhile, the RLR vehicle attempted to negotiate a red light at the same intersection. In order to avoid the collision, the driver had to decelerate.

Fig 1 showed the specific RLR crash scenario designed at the experimental intersection in the experiment. TTC sensor was selected to calculate the simulator's approaching time to the conflict point in real time and to trigger the RLR vehicle, proximity sensor was used to trigger auditory warning. The RLR vehicle was triggered to cross the intersection with a velocity of 20 m/s at point A by the TTC sensor when the simulator was 7 s away from the conflict point at point B. An auditory warning will be not be published to alert the driver until the approaching time to the experimental intersection reached a pre-designed releasing time.

The main factor in this experiment was releasing time of ICWS warnings. In this study, the latest releasing time of warning should at least allow drivers to react to the warning and the earliest releasing time of warning could not increase user's distrust of warning, thus, seven kinds of releasing time of warning that varied from 2.5 s to 5.5 s before collision

occurred with 0.5 s interval was set. With no warning as control group, eight experimental intersections were designed for each participant, containing seven potential collision event paired with different releasing time of ICWS warnings and the control group.

In the experiment, the eight experimental intersection were randomly distributed in five similar driving scene. Each driving scene would last about 5 min to complete, which consisted of two-way two-lane urban roadway that 3.5 m wide and the speed limit was 80 km/h. Six signalized intersections were located every 800-1200m in a driving scene, and one or two experimental intersection was randomly assigned to the six similar intersections in each driving scene, so that drivers could not predict the conflict event according to location or the temporal order in memory during the experiment operation. Therefore, all of ICWS warnings were presented in the driving simulations, and individual drivers only encountered each ICWS warning once and only drove each scene and scenario once.

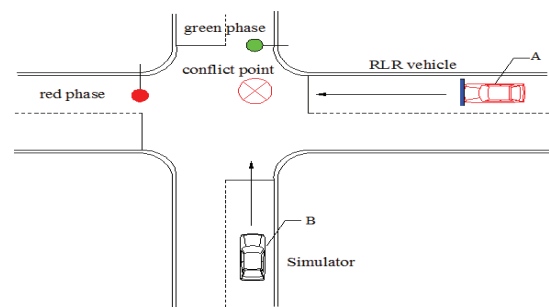


Fig. 1. RLR crash designed at the experimental intersection

C. Procedure and data analysis

For the study, all participants first provided their personal information, including sex, age, licensure status, and experience with video games. Before the formal experiment, all participants were given experimental instruction for the driving task and the subjects were instructed in how to operate the simulator, including driving straightly, accelerating, decelerating, left/right turning, and then they were given a 10-15min driving practice to adapt the operation of simulator. During each scenario, the drivers could take a rest of 10 min. Besides, the drivers could quit the experiment at any time if they feel any motion sickness, and their unfinished experiments data would be deleted from the dataset. At the end of the experiment, the drivers had to finish a questionnaire about the subjective effectiveness to ICWS warnings by participants themselves. Besides, they could get financial compensation for the time taken to complete the experiment. In this study, the driving speed and distance to the conflict point were used to evaluate driving behaviors when encountered potential collisions, which could be obtained by the driving simulator in real-time. Paired T-test was utilized to examine group differences among speed and location in different ICWS warnings conditions, especially on the different releasing time of warning. Analysis of variance (ANOVA) was also used to examine differences among

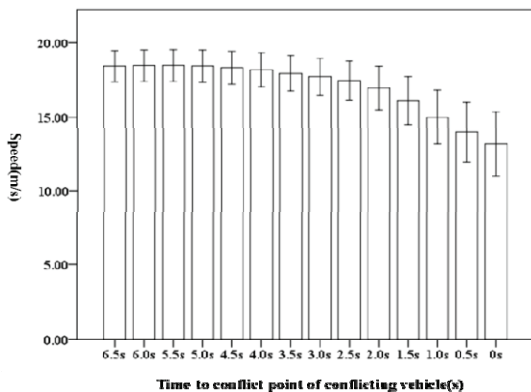
distance to conflict point (when the deceleration rate reaches the maximum) in different releasing time of warning. In each analysis 0.05 was used as the minimum level of significance.

III. RESULTS AND DISCUSSION

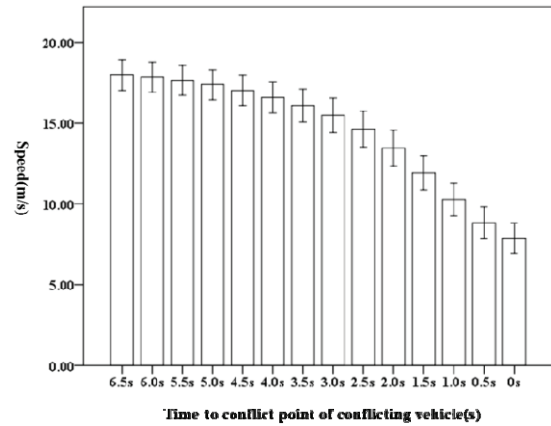
A. Driving Speed

When drivers were gradually approaching signalized intersection, they would be more cautious to adjust the speed. As is shown in Figure 2, during the early stage of potential collision events (6.5 s to 5.5 s to conflict point), drivers were approaching the intersection with no obvious speed change, which might be caused by a low traffic flow and green phase ahead. During the middle stage of releasing warning (5.5 s to 0 s to conflict point), drivers started to decelerate at different time distance to the conflict point under different releasing time of warning. Additionally, the last part of this stage can indicate results of collision avoidance measures under different releasing time of warning.

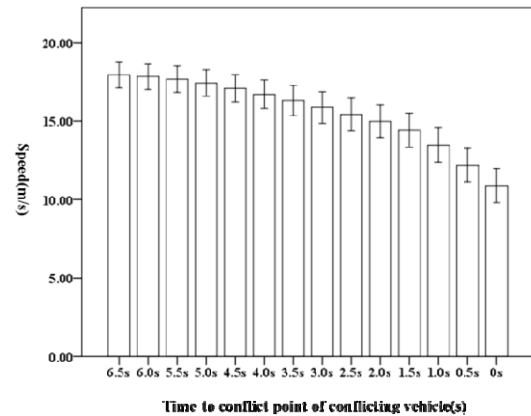
The speed change trend of no warning condition was shown in Figure 2-a, the speed slowed down slightly until time distance to conflict point was 1.5 s, when the drivers intensify the strength on brake pedal. However, such behavior was found to correspond with a high crash rate (44.1%). But for warning conditions, the trend of decelerating was more obvious than that of no warning, which indicated that warning could reduce speed when drivers crossed an intersection and indirectly reflected that warning system can enhance driving safety. Similar results were also identified by previous studies [1, 12]. When ICWS warning was provided 2.5 s (see Figure 2-b) before collision, the crash rate was second high (29.4%) and there was no violent deceleration. For warning provided 3.0 s (see Figure 2-c) before collision, sharp deceleration appeared at 2.5 s before the conflict point. For warning provided 3.5 s (see Figure 2-d) before collision, more obvious deceleration was adopted at the time distance of 1.5 s to the conflict point. For warning provided 4.0 s (see Figure 2-e) before collision, sharp deceleration was adopted at the time distance of 2.5 s to the conflict point, and the same situation occurred when the warning was provided 4.5 s (see Figure 2-f) before collision. For warning provided 5.0 s (see Figure 2-g) before collision, violent decelerations were adopted at 3.5 s before the conflict point, which was nearly the same as the condition in releasing time of 5.5 s (see Figure 2-h).



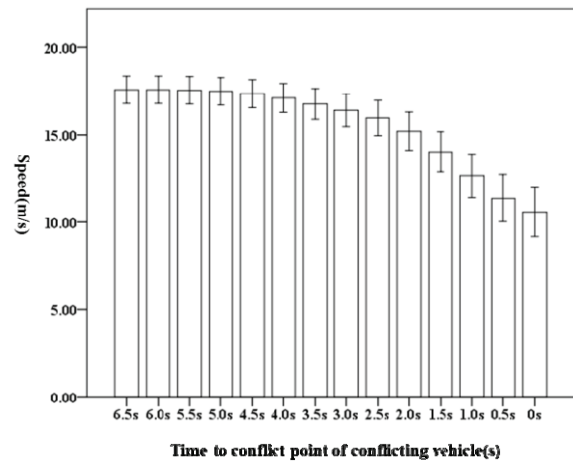
(a) Releasing time of warning = NA



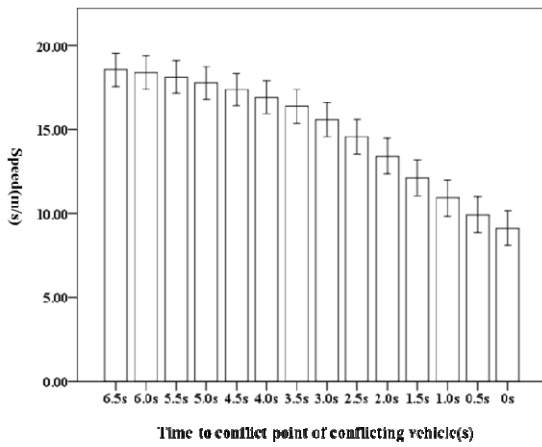
(b) Releasing time of warning = 2.5 s



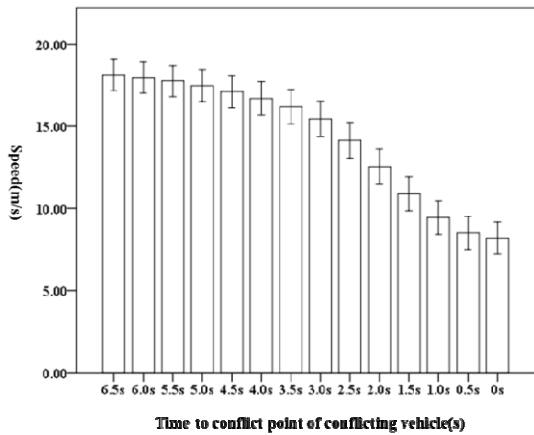
(c) Releasing time of warning = 3.0 s



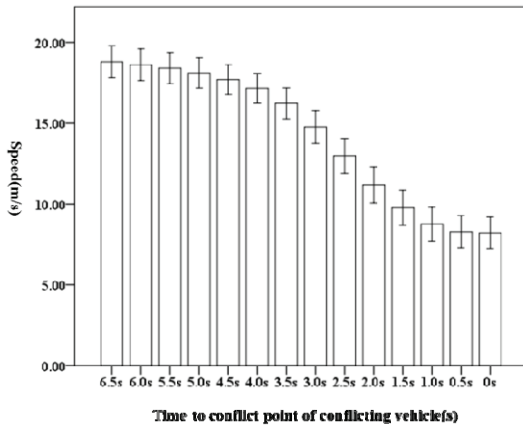
(d) Releasing time of warning = 3.5 s



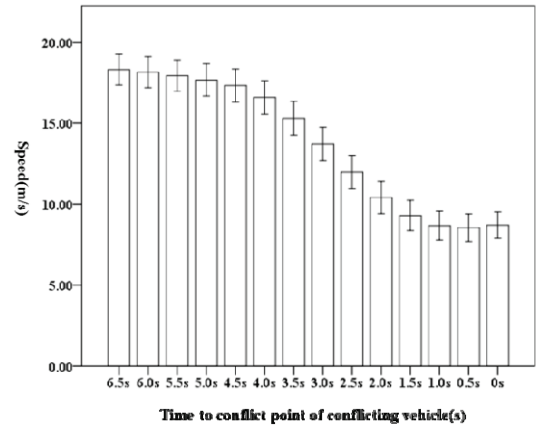
(e) Releasing time of warning = 4.0 s



(f) Releasing time of warning = 4.5 s



(g) Releasing time of warning = 5.0 s



(h) Releasing time of warning = 5.5 s

Fig. 2. Mean speed with different ICWS warning conditions

TABLE I. THE DESCRIPTIVE STATISTIC OF CAR-FOLLOWING REACTION TIME BY SCENARIOS

Mean differences between	Time to conflict point of conflicting vehicle(s)				
	6.5s	6.0s	5.5s	5.0s	4.5s
NA & 2.5s	-0.24	0.09	0.26	0.38	0.46
NA & 3.0s	-0.09	0.60	0.95	1.28	1.59
NA & 3.5s	-0.24	0.11	0.12	0.13	0.13
NA & 4.0s	-0.02	0.26	0.59	0.93	1.26
NA & 4.5s	0.06	0.53	0.77	0.98	1.18
NA & 5.0s	-1.54	-0.98	-0.76	-0.41	0.13
NA & 5.5s	0.44	0.92	1.18	-0.41	1.62
Mean differences between	Time to conflict point of conflicting vehicle(s)				
	4.0s	3.5s	3.0s	2.5s	2.0s
NA & 2.5s	0.50	0.53	0.64	0.78	0.68
NA & 3.0s	1.90	2.17*	2.44*	2.77*	2.92*
NA & 3.5s	0.14	0.13	0.13	0.12	0.11
NA & 4.0s	1.63	2.01	2.36*	2.71*	2.82*
NA & 4.5s	1.45	1.74	2.56*	4.18*	5.71*
NA & 5.0s	0.91	2.08*	3.87*	5.64*	7.00*
NA & 5.5s	2.32*	3.61*	5.24*	6.94*	8.08*
Mean differences between	Time to conflict point of conflicting vehicle(s)				
	1.5s	1.0s	0.5s	0s	--
NA & 2.5s	0.39	0.25	0.45	0.82	--
NA & 3.0s	3.25*	3.71*	4.01*	4.12*	--
NA & 3.5s	0.06	-0.01	-0.02	0.06	--
NA & 4.0s	2.81*	2.74*	2.67*	2.82*	--
NA & 4.5s	6.62*	6.71*	6.00*	5.13*	--
NA & 5.0s	7.24*	6.28*	5.02*	4.09*	--
NA & 5.5s	8.01*	6.89*	5.44*	5.13*	--

*p<0.05

Table 1 has showed comparison results of speed change trend between no warning and warning with different releasing time, and the t-value of Matched T-test is also listed in the Table 1. As is shown in Table 1, no significant difference existed in speed trend between no warning and 2.5 s warning conditions, as well as no warning and 3.5 s warning conditions, which indicated that late warning provided no help for drivers to adjust speed. However, an interesting result was found that in the conditions of 3.0 s warning, 4.0 s warning and 4.5 s warning, time distance to the conflict point that was less than 3.0 s was statistically different from that in no

warning, which indicated that the effectiveness of auditory warning starts to reflect from 3.0 s before collision. For 5.0 s warning, time distance to conflict point that was less than 3.5 s was statistically different from that in no warning, which indicated that effectiveness of auditory warning starts to reflect from 3.5 s before collision. For 5.5 s warning, time distance to conflict point that was less than 4.0 s was statistically different from that in no warning, which indicated that effectiveness of auditory warning starts to reflect from 4.0 s before collision. Above all, the appropriate time of violent deceleration for different releasing time of warning was about 3.0 s to 4.0 s, which might also be a proper time to provide warning. With regard to the warning timing, previous studies [1, 18] had consistently found that early warning was more effective to increase the effectiveness of CWS. The current result predicted that early warning had a clear positive effect of reducing speed, thus drivers would be more quick to take braking for the critical event in order to avoid a collision. Also in late warnings, the time that drivers step on the brake pedal strongly was later than that in early warnings, which might cause uncomfortable deceleration and more serious consequence if the collision was inevitable.

B. Distance to conflict point with maximum deceleration

At the signalized intersection, if a RLR crash develops, the conflict point or area could be determined. When drivers pressed the brake pedal to reach the maximum value, the distance to the conflict point was calculated and analyzed by ANOVA, which was an important indicator to show safe distance to a potential collision.

Table 2 had listed the drivers' distance to conflict point at maximum deceleration rate under different warning releasing time. Experimental results indicated that the mean distance without warning information (33.24 m) was smaller than those of with warning information (36.57 m, 44.44 m, 36.92 m, 50.07 m, 50.02 m, 56.26 m, 59.49 m), indicating that the different ICWS releasing time of warnings made a significant difference on the driver's safety distance ($p < 0.05$) to the conflict vehicle. The results showed that the warning with earlier releasing time could increase distance to conflict point when the deceleration rate reached maximum. The possible reason was that with earlier warning, drivers could respond to a potential collision earlier so that they could reserve more space to eliminate the emergent event or other risk.

TABLE II. THE DESCRIPTIVE STATISTIC OF CAR-FOLLOWING REACTION TIME BY VOCATION

Variables	Classification	Distance to conflict point with maximum deceleration		F-value
		Mean(m)	S.D.	
Releasing time of ICWS warning	No warning(NA)	33.24	40.75	7.464*
	2.5 s	36.57	32.06	
	3.0 s	44.44	28.10	
	3.5 s	36.92	30.12	
	4.0 s	50.07	28.95	
	4.5 s	50.52	24.82	
	5.0 s	56.26	29.09	
	5.5 s	59.49	15.97	

* $p < 0.05$

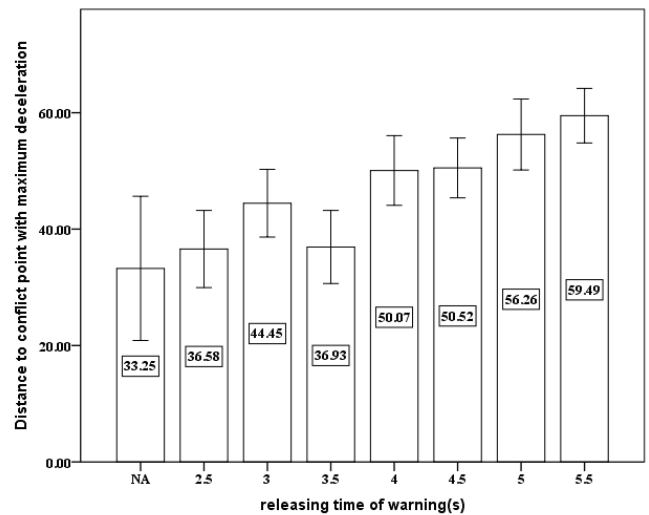


Fig. 3. Mean distance to conflict point with maximum deceleration with different ICWS warning conditions.

IV. CONCLUSIONS

This study mainly investigated drivers' speed trend and distance to conflict point at maximum deceleration rate under different ICWS releasing time of warning conditions. Previous research had showned that, firstly, compared to no warning, drivers slowed down to make a decision to brake after listening the auditory warning, which demonstrated that releasing an ICWS warning could be expected to provide some help in reducing driving speed to the hazard event. That was to say, warning information was presented to impact the adjustment of speed to respond appropriately to the RLR collision situation, and the results presented a trend of earlier violent deceleration and longer distance to conflict point when releasing warning in the critical intersection situation.

According to the results of driving speed and distance to the conflict point under different releasing time of warning conditions, warning should be released at least 3.0 s before a collision at a signalized intersection, and considering the time of application of violent deceleration for different releasing time of warning, 3.0 s to 4.0 s before a collision occurred might be a proper time to provide warning. Additionally, the time of the maximum deceleration rate appeared earlier as the time of warning was released earlier, and drivers would reserve more room to clear impending collision to response more safely and comfortable with the earlier warning. Overall, the results in this study have a mutual promotion with the existing body of literature in revealing that how the different releasing time of warnings affects drivers' performance at the critical intersections.

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